

# Yong Guk Kang

Research Professor, Imaging Intelligence Laboratory, School of Mechanical Engineering, Seoul National University

Optical systems and computational imaging researcher focused on physics-informed simulation, inverse optical design, lensless imaging, and machine-learning assisted image reconstruction. I translate research problems into imaging system models, reconstruction pipelines, and experimental strategies for optical and biomedical applications.

ygkang@snu.ac.kr | <https://github.com/kygwow>

## RESEARCH FOCUS

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Computational imaging and inverse optical design; Lensless and phase-coded imaging systems; AI-assisted image reconstruction and virtual staining; Quantitative phase, interferometric, and coherent imaging; Physics-informed modeling of optical imaging systems

## SELECTED CAPABILITIES

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**Computational imaging:** Wave-optics simulation, Inverse problems and image reconstruction, Machine learning assisted optical design and reconstruction

**Optical imaging:** Quantitative phase imaging, Interferometric microscopy, Optical coherence microscopy, Nonlinear microscopy

**Tools:** MATLAB, Python, PyTorch, ZEMAX, Fusion 360, ImageJ

## TECHNICAL SCOPE

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**Computational imaging:** Wave optics simulation, Lensless imaging, Inverse design, Optical image reconstruction, Physics-based noise modeling, Interferometric imaging, Image deconvolution, Virtual staining

**Optical imaging modalities:** Quantitative phase imaging, Interferometric microscopy, Nonlinear microscopy, Optical coherence tomography, Optical coherence microscopy, Off-axis holographic microscopy

**Computational tools:** MATLAB, Python, PyTorch, ZEMAX, ImageJ

**Experimental implementation:** Optical system prototyping, Microfabrication workflows, 3D CAD and rapid prototyping

## EXPERIENCE

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**Research Professor**, Seoul National University, School of Mechanical Engineering (2025-04-present)

- Developing computational imaging frameworks that combine optical system modeling, inverse design, and data-driven reconstruction.
- Investigating lensless and phase-coded imaging strategies for high-dynamic-range and information-rich optical measurements.
- Applying physics-informed simulation and machine learning to jointly reason about optical encoding and image reconstruction.

**Researcher**, Yonsei University, School of Electrical and Electronic Engineering (2024-04-2025-02)

- Contributed to computational imaging and lensless imaging research in the Imaging Intelligence Laboratory.
- Worked on phase-coded optical systems, wave-optics modeling, and reconstruction-oriented imaging methods.
- Collaborated on compact optical imaging platforms and biomedical imaging applications.

**Postdoctoral Researcher**, Korea University, School of Biomedical Engineering (2022-09-2024-03)

- Advanced reflection phase microscopy for efficient 3D quantitative imaging with reduced axial scanning.
- Developed computational refocusing and field synthesis strategies for reflection phase measurements.
- Integrated optical modeling and experimental validation for interferometric biomedical imaging.

**Research Lecturer**, Korea University, School of Medicine (2020-10-2022-08)

- Developed optical coherence microscopy approaches for quantifying cell-ECM interactions and intracellular dynamics.

- Applied temporal-dynamics contrast imaging to live-cell and tissue-scale biomedical measurements.
- Connected optical imaging measurements with quantitative analysis of mechanical interactions in biological matrices.

## EDUCATION

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**Ph.D. in Biomedical Engineering**, Korea University

**B.S. in Biomedical Engineering**, Korea University

## PROJECTS

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**Lensless and HDR computational imaging.** Computational imaging research on lensless and phase-coded acquisition, with emphasis on optical encoding, inverse design, and robust reconstruction under challenging measurement conditions.

**Reflection phase microscopy.** Quantitative reflection phase microscopy for efficient 3D surface and biomedical imaging, combining interferometric measurement design with computational refocusing and field synthesis.

**Optical coherence microscopy for cell-ECM dynamics.** Optical coherence microscopy and temporal-dynamics contrast imaging for studying cell, tissue, and extracellular matrix interactions.

**Live-cell nonlinear microscopy for matrix deformation analysis.** Nonlinear microscopy and quantitative image analysis for studying collagen matrix deformation and cell-ECM mechanical interactions.

## SELECTED JOURNAL ARTICLES

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Geometry-aware phase compensation for sampling-efficient angular spectrum method, *Optics Express* (2026)

Wide-area topography using reflection phase microscopy for surface inspection, *Optics Express* (2025)

Semi-supervised virtual staining using learned-illumination Fourier ptychography for high-speed label-free histopathology, *Journal of Physics: Photonics* (2025)

Cardiac fibroblast-mediated ECM remodeling regulates maturation in an in vitro 3D engineered cardiac tissue, *Journal of Tissue Engineering* (2025)

Laser-tissue interaction simulation considering skin-specific data to predict photothermal damage lesions during laser irradiation, *Journal of Computational Design and Engineering* (2023)

Three-dimensional imaging in reflection phase microscopy with minimal axial scanning, *Optics Express* (2023)

Optical coherence microscopy with a split-spectrum image reconstruction method for temporal-dynamics contrast-based imaging of intracellular motility, *Biomedical Optics Express* (2023)

Development of a 3-D Physical Dynamics Monitoring System Using OCM with DVC for Quantification of Sprouting Endothelial Cells Interacting with a Collagen Matrix, *Materials* (2020)

Reflection Phase Microscopy by Successive Accumulation of Interferograms, *ACS Photonics* (2019)

Quantification of focal adhesion dynamics of cell movement based on cell-induced collagen matrix deformation using second-harmonic generation microscopy, *Journal of Biomedical Optics* (2018)

Single-shot digital holographic microscopy for quantifying a spatially-resolved Jones matrix of biological specimens, *Optics Express* (2016)

## SELECTED CONFERENCE PAPERS

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High-speed virtual re-staining via near-infrared quantitative phase imaging, *Proceedings of SPIE Advanced Biophotonics Conference* (2026)

A compact, bite-down lensless imager for 3D fluorescence mapping of occlusal dental plaque, *Proceedings of SPIE* (2026)

Non-Paraxial Layered Diffraction Simulator for Inverse Design of Lensless Imaging Optics, *Optica Imaging Congress 2025* (2025)

## PATENTS AND RECOGNITION

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**Funding:** Creative and Challenging Research Base Support (2021R111A1A01048370)

**Fellowship:** Global Ph.D. Fellowship (2015H1A2A1030048)

**Patent:** Head mount system for providing surgery support image (US 11,356,651)

**Patent:** Reflection phase microscopy system for 3D image acquisition (10-2024-0081557)

**Patent:** Phase mask fabrication method based on grayscale lithography (10-2024-0161144)